

Isolated Processes and App Zygote Preloading on Android

Isolated Processes

Isolated processes on Android are an OS level concept that uses [SELinux permissions](#) to sandbox processes. Isolated processes are separate from Fission, the Gecko technology that separates rendering processes based on origin. A service that is using isolated processes will have

`android:isolatedProcess="true"` in the manifest.

Isolated processes have been supported since Android 4.1 (level 16). To use `bindIsolatedService`, an API level of Android 10 (level 29) is required. `android.os.Process.isIsolated()` is available since Android 4.1 (level 16), but a hidden API until Android 9 (level 28).

There are many differences between isolated processes and regular processes.

Some important differences include:

- Isolated processes are not allowed to start or bind most other components (e.g. services, activities)
- The UID of the isolated process is different from the parent process, so it does not share permissions, identity, or UID-based file access
- Isolated processes do not inherit the app's runtime permissions and must rely on IPC to a non-isolated process for permission-gated operations
- Access to system services, app state, and filesystem paths is more restricted due to tighter SELinux confinement
- Many system calls and kernel interactions are restricted compared to regular app processes

Some links to external documentation:

- [Video overview of isolated processes](#)
- [SELinux Contexts](#)
- [Android SE Policy](#)
- [SELinux Policy Rules for isolated processes on Android](#)

How to run isolated process mode

GeckoView Test Runner Locally

`ac_add_options --enable-isolated-process` (Should apply to junit, supported mochitests, xpcshell, wpt, and reftests.)

GeckoView Example

Hardcode `mIsolatedProcess` to true in `GeckoRuntimeSettings.java`.

CI

Currently in CI as a build type of `debug-isolated-process`. It is in the process of transitioning to a variant of `geckoview-isolated-process`.

Fenix

Isolated processes may be tested by enabling it in Fenix's secret settings and restarting the app.

Note: If app zygote preloading is also enabled, those settings will take precedence. Also, the GV flags will not apply because Fenix has control of the setting.

How to confirm isolated processes

```
# GeckoView Test Runner
adb shell "ps -Z -A | grep -i geckoview.test"

# Fenix
adb shell "ps -Z -A | grep -i fenix"

# Note: The exact full process name will vary based on release type, but
should still be found using "fenix". e.g. beta, debug
```

Example Isolated Process

✓ Yes, isolated process:

```
u:r:isolated_app:s0:c512,c768 u0_i9795 19244 5586 17883664 217912
do_epoll_wait 0 S
org.mozilla.geckoview_example:isolatedTab0:org.mozilla.gecko.process.GeckoCh
    ^^^^^^^^^^^^^^^^^      ^^^^^^^^^
    ^^^^^^^^^^^^^^^^^
    (isolated_app)          (u0_i prefix)
    (isolatedTab)
```



Key indicators: `isolated_app` type, `u0_i` prefix in UID, `isolatedTab` in the process name

Only the tab processes will be stated as isolated. Other processes won't be isolated.

Example Regular Process

✗ Not isolated process:

```
u:r:untrusted_app:s0:c75,c257,c512,c768 u0_a331 19936 5586 17833968 170608
do_epoll_wait 0 S org.mozilla.firefox_beta:tab30
    ^^^^^^^^^^^^^^^^^      ^^^^^^^^^
    ^^^
    (untrusted_app)          (u0_a prefix)
    (tab)
```

Key indicators: `untrusted_app` type, `u0_a` prefix in UID, and `tab` in process name (not `isolatedTab`)

App Zygote Preloading

For Firefox on Android, [app zygote preloading](#) creates a zygote process with libraries preloaded to help increase efficiency when launching new tab processes. The libraries selected are defined in [ZygotePreload.java](#).

Isolated processes working as expected is a prerequisite for app zygote preloading. We automatically enable isolated processes for the app zygote case.

App zygote preloading requires Android 10 (level 29) or higher.

How to run app zygote preloading

GeckoView Test Runner Locally

`ac_add_options --enable-isolated-zygote-process` (Should apply to junit, supported mochitests, xpcshell, wpt, and reftests.)

GeckoView Example

Hardcode `mAppZygoteProcess` to true in `GeckoRuntimeSettings.java`.

CI

`geckoview-zygote` variant

Fenix

App zygote preloading can be tested by enabling it in Fenix's secret settings and restarting the app.

Note: The GV flags will not apply because Fenix has control of the setting.

How to confirm app zygote preloading

```
# GeckoView Test Runner
adb shell "ps -Z -A | grep -i geckoview.test"

# Fenix
adb shell "ps -Z -A | grep -i fenix"
```

Example App Zygote Preloading

✓ Yes, app zygote preloading process:

```
u:r:app_zygote:s0:c512,c768 u0_a212 6475 350 15686420 180720 do_sys_poll 0
S org.mozilla.fenix.debug_zygote
  ^^^^^^^^^^^
  (app_zygote)
```

Key indicator: `app_zygote` type

✓ Yes, isolated process that launched via app zygote:

```
u:r:isolated_app:s0:c512,c768 u0_i0 6525 6475 18017956 315748
do_epoll_wait 0 S
org.mozilla.fenix.debug:zygoteTab:org.mozilla.gecko.process.GeckoChildProcess
      ^^^^^^^^^^^^^^^^^      ^^^^^
^^^^^^^^^^
      (isolated_app)      (u0_i)
      (zygoteTab)
```



Key indicators: `isolated_app` type, `u0_i` prefix in UID, and `zygoteTab` in the process name